CSCI 271: Assignment 2
Name: CHENKAI NIU Points: _____ / 25
10 multiple-choice questions covering variables, data types, arithmetic, and input/output from Lesson 2. Circle or check the box next to your answer.

1. What does declaring a variable actually do?
☐ A) It immediately sets the value to 0
X B) It reserves a named location in memory with a specific type
☐ C) It prints the variable's name to the screen
☐ D) It deletes any variable with the same name

2. Which data type should you use to store a single letter grade, like 'A'?
☐ A) int
☐ B) string
X C) char
☐ D) double

3. What is the value of int result = 9 / 2;?
☐ A) 4.5
X B) 4
☐ C) 5
☐ D) It causes a compiler error

4. What does cout << 2 + 3 * 4; print?
☐ A) 20
☐ B) 9
X C) 14
☐ D) 24

5. What's the difference between << and >>?
☐ A) << reads input, >> sends output
X B) << sends output, >> reads input
☐ C) They do the same thing and can be used interchangeably
☐ D) << is only for numbers, >> is only for text

6. What does the escape sequence \t do inside a string?
☐ A) Starts a new line
☐ B) Prints a literal backslash
X C) Inserts a horizontal tab
☐ D) Deletes the previous character

7. Given int x = 5; followed by x = 8;, what happens to the original value 5?
☐ A) x now stores both 5 and 8
X B) x now stores only 8; the 5 is gone

☐ C) The program produces a compiler error
☐ D) x still stores 5 until the program ends

8. What happens when you write int x = 3.5;?
☐ A) It rounds to 4
☐ B) It causes a compiler error
X C) It stores 3, silently dropping the .5
☐ D) It stores 3.5 exactly, same as a double would

9. In the line int total = 5;, which word is an identifier rather than a keyword?
☐ A) int
X B) total
☐ C) double
☐ D) char

10. Why is x = x + 1; valid in C++, even though it looks wrong in math class?
☐ A) It's a math error — x can never equal x + 1
X B) It updates x by taking its current value, adding 1, and storing the result back in x
☐ C) It compares x to x + 1 and returns true or false
☐ D) It only works if x was never given a value before

Mini Lab: Try It Yourself
// Your task:
// 1. Ask the user to enter their name and age
// 2. Print a greeting using both values

// Example output:
// Hello, I am Choco! I am 1 year old.

Write the C++ program (explain all the lines) and add the screenshot of your output: